

AWG or VSG. A user can add impairments and verify DUT performance with sea clutter model impairments on signals.

Propsim is a radio channel emulator to simulate channel propagation effects on signals and test DUT injected mode in a lab setup.

Radar cross section measurement system measures the cross-section of the ship.

Some other test and measurement instruments are base station analysers (for example, BTS Master), Bluetooth and WLAN testers, cable and antenna analysers (for example, Site Master), channel emulator conformance test systems, handset one box testers, radio communication analysers, land mobile radios (for example, LMR Master), passive intermodulation analysers (for example, PIM Master), power meters and sensors, shield box signal generators, signalling testers (base station simulators) and signal analysers.

Devices in the market. Madhukar Tripathi, head - marcom and optical business, Anritsu India, says "The ultra-portable spectrum analyser MS2760A utilises non-linear transmission line technology and offers dynamic range, sweep speed and amplitude accuracy. Its size enables direct connection to almost any DUT, eliminating the need for lossy, expensive cables or antennae. It provides continuous coverage from 9kHz to 110GHz, and is great for millimetre-wave applications like 5G, 802.11ad, satellite communications, automotive radars and more. It measures channel power, adjacent channel power and occupied bandwidth."

Anritsu's MS2850A provides maximum analysis bandwidth of 1GHz and frequency range of 9kHz to either 32GHz or 44.5GHz. It can capture up to 24 seconds at 100MHz span using 32GB memory. Signal data can be analysed for efficient R&D of next-generation standards prior to release.

PCIe/ISB3.0 enables high-speed transfer of large captured data.

Radio communication test station MT8000A by Anritsu provides all-in-one support for RF measurements and protocol tests in sub-6GHz and millimetre-wave bands. After



Fig. 2: Propsim F64 5G channel emulation solution F8800A (Credit: Keysight Technologies)

combining with an OTA chamber, it enables both millimetre-wave band RF measurements and beamforming tests specified by 3GPP.

Emulators for creating test environments

Propsim channel emulation solution is a radio channel emulator that enables recreating wireless channel propagation effects in a controlled lab environment. It is a one-box solution for performing a realistic and accurate emulation of all typical radio channel propagation effects, such as multi-path propagation, fast fading, dynamic delays, attenuation, noise, interference and shadowing. It enables real-time performance testing of devices and base stations. The solution emulates impairments of complex real-world radio channel conditions for accurate testing with dynamic multi-path propagation, range path-loss and blocking effects, Doppler from mobility and multi-paths, and noise and synchronous programmable interference.

Imran Baig, senior application engineer, Keysight Technologies India, says, "Propsim fulfils the requirements of both present and future wireless communication systems. It provides feature sets for radio interface testing of current cellular communication systems, such as LTE/LTE-A, WLAN, WCDMA, GSM, TD-SCDMA, etc.

"Additionally, Propsim enables testing of next-generation communication systems, such as 5G and VHF communication systems operating at 30MHz - 300MHz frequency. It includes pre-stored standard channel emulations (emulation connection diagram with channel models according to 3GPP standards).

"For basic use, the user only needs to load the emulation and adjust the power levels of Propsim. For advanced use, all versatile channel modelling tools for creating user-defined channel models and for modifying predefined standard channel models are available."

Propsim F64 5G channel emulation solution F8800A addresses diverse test needs in 5G, 4G and WLAN 802.11ax.

Field-to-lab virtual drive testing emulates impairments of complex 3D real-world radio channel conditions to support user-defined 3D spatial scenarios and dynamic modelling of movement, arbitrary and complex test scenarios, multi-frequency and multi-RAT HetNet test scenarios, device-to-device (D2D) supporting Internet of Things (IoT) and vehicle-to-everything (V2X) scenarios, 3D antenna beam pattern model embedding, antenna array and antenna library tool for modelling arrays and beams.

Challenges and improvement areas

Communication signal standards are changing at a fast rate. Also, communication bands are overlapped into microwave frequency bands. Hence, testing DUT performance against only a particular test signal cannot give true performance. It is necessary to simulate all kinds of signals and inject these into the systems.

Latest communication radios are software-defined with fast-hopping signals up to 1000 hops per second and beyond. It would be difficult to jam these signals with wide-band jamming techniques. Research is on for developing follow-on jamming techniques, where the jammer system follows each hop of the radio. So, faster switching source and receiver looks challenging. **EFY**